

# HANYEOL RYU

Pusan National University | Email: [9lo522@pusan.ac.kr](mailto:9lo522@pusan.ac.kr) | Web: <https://hyrecon.github.io/>

## RESEARCH INTERESTS

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- Low-Power Wireless Communications (e.g. Backscatter Communications)
- Channel Modeling & Estimation
- AI/ML for Intelligent Wireless Communications
- Self-Sustainable mmWave Communication and Sensing Systems

## EDUCATION

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### Pusan National University

- *Master of Science (M.S.) in Electrical and Electronics Engineering* 2026 – Present
- *Bachelor of Science (B.S.) in Electronics Engineering* 2021 – 2026

## RESEARCH EXPERIENCE

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### Pusan National University

- *Research Assistant (PI: Dr. Sangkil Kim)* Jan 2024 – Present

## PROFESSIONAL ACTIVITIES

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### Technical Program Committee (TPC) Reviewer

- IEEE ISPACS 2025

### Journal Reviewer

- IEEE Internet of Things Journal (IoT-J) 2026

## PUBLICATIONS

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1. **Hanyeol Ryu**, Nohgyeom Ha, Sangkil Kim, “Transformer-Hypernetwork-Controlled Deep-Unfolded Phase-Aware Channel Estimation Refinement for Phase-Drift-Robust Backscatter Links,” *arXiv preprint*, 2026. (DOI: [10.48850/arXiv.2606.31400](https://doi.org/10.48850/arXiv.2606.31400)).
2. **Hanyeol Ryu**, Sangkil Kim, “Deep Learning-Aided Phase-Aware Channel Estimation for Error-Floor Suppression over Phase-Drifting Backscatter Links,” *IEEE International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS)*, Accepted, 2026.
3. Nohgyeom Ha, **Hanyeol Ryu**, Hyunmin Jeong, Hoyong Kim, Gyoungdeuk Kim, Apostolos Georgiadis, Manos M. Tentzeris, Sangkil Kim, “Integrated Sensing and Backscatter Communication: Toward Self-Sustainable Autonomous Internet of Things Networks,” *IEEE*

*Microwave Magazine*, 2026. (DOI: [10.1109/MMM.2026.3717878](https://doi.org/10.1109/MMM.2026.3717878))

4. **Hanyeol Ryu**, Sangkil Kim, “Ultralow-Power Monostatic Backscatter Platform With Phase-Aware Channel Estimation and System-Level Validation,” *IEEE Internet of Things Journal*, 2026. (DOI: [10.1109/JIOT.2026.3696951](https://doi.org/10.1109/JIOT.2026.3696951))
5. **Hanyeol Ryu**, Sangkil Kim, “High-Isolation Yagi-Uda Antenna Array using Split-Ring Resonators for Practical Backscatter System,” *IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting (AP-S/URSI)*, 2025. (DOI: [10.1109/AP-S/CNC-USNC-URSI55537.2025.11266818](https://doi.org/10.1109/AP-S/CNC-USNC-URSI55537.2025.11266818))
6. Jeonghyeon Choi<sup>1</sup>, **Hanyeol Ryu**<sup>1</sup>, Sangkil Kim, “Improved Backscattering Communication Range Using Reflection Amplifiers,” *The Journal of Korean Institute of Electromagnetic Engineering and Science*, 2025. (DOI: [10.5515/KJKIEES.2025.36.3.266](https://doi.org/10.5515/KJKIEES.2025.36.3.266))

## PATENT

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- Korean Patent Application, Allowed April 2026. “Apparatus and Method for Phase-Aware Channel Estimation in Monostatic Backscatter Communication Systems.”  
**Hanyeol Ryu**, Sangkil Kim.

## CONFERENCE PRESENTATIONS

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Title: *Deep Learning-Aided Phase-Aware Channel Estimation for Error-Floor Suppression over Phase-Drifting Backscatter Links*

- IEEE ISPACS Oct 2026

Title: *Ultralow-Power Backscatter Platform With Phase-Aware Channel Estimation*

- The Korean Institute of Electromagnetic Engineering and Science (KIEES) Feb 2026

Title: *High-Isolation Yagi-Uda Antenna Array using Split-Ring Resonators for Practical Backscatter System*

- IEEE AP-S/URSI Jul 2025

Title: *Improved Backscattering Communication Range Using Reflection Amplifiers*

- KIEES Feb 2025

## HONORS & AWARDS

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- **Excellence Award (2<sup>nd</sup> Place)**, Hackathon, Pusan National University Aug 2023
- **Excellence Award (2<sup>nd</sup> Place)**, Capstone Design Project, Pusan National University Nov 2025

## SERVICE & EXTRACURRICULAR ACTIVITIES

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### Pusan National University

- Treasurer, Department Student Council Dec 2021 – Dec 2022

- Student Ambassador, Pusan National University Apr 2021 – Feb 2023
- Volunteer, Overseas Volunteering Program (3D Printer & Arduino) — Thailand Dec 2022 – Feb 2023

### Military Service

#### **Republic of Korea Marine Corps (ROKMC)**

- Squad Leader, ROKMC Force Reconnaissance (Force Recon) 2017 – 2018
- 2<sup>nd</sup> Place, ROKMC Force Recon Qualification Course 2017

### **SKILLS & TECHNIQUES**

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- **Programming:** MATLAB, Python, C++
- **AI/ML:** PyTorch, TensorFlow/Keras
- **RF/EM Simulation:** Ansys HFSS, Advanced Design System (ADS)
- **Software-Defined Radio:** GNU Radio
- **Hardware Design:** OrCAD PCB Editor, OrCAD Capture
- **3D Modeling:** Rhino 7